

Food Analytics – A literature review and ways forward

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Abstract— This research examines the potential use of modern technologies such as big data, data science, artificial intelligence, and machine learning, which have penetrated several aspects of our lives, to address food concerns and problems, forming the nowadays called food analytics. We discuss the potential use of such technologies in relation to food problems and shortages. We analyze the opportunities and challenges associated with the use of such technological advancements and the potential benefits for the global food system. We devise an articulated research agenda with future directions for the application of big data, data science, artificial intelligence, and machine learning to food ecosystem.

Keywords—AI, big data, data science, food analytics, research agenda

I. INTRODUCTION

With the world population expected to rise dramatically over the next few decades, and in the face of extra stress caused by climate change and urbanization, one of the most pressing challenges that the world faces today is securing basic resources such as food, energy, and water [1]. A food system can be broadly defined as encompassing "all the elements (environment, people, inputs, processes, infrastructures, institutions, etc.) and activities related to the production, processing, distribution, preparation, and consumption of food, and the outputs of these activities, including socio-economic and environmental outcomes" [2].

By engaging in activities such as deforestation, urbanization, transportation, and modifying the landform and vegetation cover, humans alter the ecological, hydrological, and thermal features of their surroundings. Consequently, there is an interdependent relationship between human and natural systems. Further, population growth, climate change, and development have intensified natural resource usage and environmental degradation to levels that jeopardize the stability and security of food, energy, and water system (FEWS) services [1]. Nowadays, the importance of data and analytics is escalating, as well as their implementation in several industries, for they enable practitioners to come up with data-driven (DD) solutions. Therefore, enhancing products and services while saving scarce resources is a win-win!

Data and analytics are of great importance to researchers, for they help them come up with data-driven decisions (DDD) [3] to address life-threatening situations. For this reason, we believe that this cross-disciplinary research would help bridge a gap, as well as serve one of the United Nations' (UN) Sustainable Development Goals (SDG)- goal number 2- "Zero

Hunger". The food industry is massively diverse and includes lots of intricate processes thus, with the help of data analytics (DA), we could help bridge a gap.

This paper aims to answer the following research question: *how could the use of big data and analytics algorithms help identify, address, or contribute to the solution of some of the food system's challenges and concerns?*

II. METHODOLOGY

To address the aim of this research, we followed a concept-centric literature review approach as outlined by Webster and Watson [4]. This approach contrasts with the author's centric approach in which the readers are usually familiar with the main topic and there are already available studies that discussed the main topic in detail. We chose the concept-centric method as it allows us to systematically synthesize the literature and enables us to create a preliminary classification on the food analytics literature.

The literature review process began with identifying core journals and conferences in food analytics. We reviewed their table of contents and manually searched for relevant articles based on title, abstract, and keywords. Additionally, we searched online databases (Scopus, Web of Science, EBSCO, PubMed, MDPI, Taylor & Francis) using specific search terms. Keywords included food, analytics, big data, AI, and IoT, with various combinations such as "food analytics," "big data for food," "AI in food," and "IoT in food analytics." Logical combinations of keywords were also employed, such as "food" AND "IoT" AND "analytics," "big data" AND "food," and "food" AND "AI."

To broaden our search for relevant studies, we conducted backward and forward citation analyses following Webster and Watson's recommendations [4]. This was necessary due to the limited initial findings. Given the emerging field of food analytics, we also included secondary resources like industrial reports and press releases. Only English-language publications were considered, with no time limitations set for this review.

III. THEORETICAL BACKGROUND

Many disciplines are now attempting to use data science (DS), which entails the use of algorithms running on top of big data (BD) following a process, to extract information and insights from structured and unstructured data and thus be able to address traditional problems, e.g., food safety, via innovative

technologies. We present theoretical background on BD, DS, artificial intelligence (AI), machine learning (ML), and finally, the internet of things (IoT) in this section so that we can use them later in the paper as a lens to review current breakthroughs in food that have resulted in what is known as food analytics.

Analytics is a multidisciplinary subject that depends on a wide range of skills and knowledge from several domains, such as BD, DS, AI, ML, and IoT, all of which work in a contextual domain and expertise. The ability to deal with and analyze diverse types of data, including structured, unstructured, and streaming data, is a critical part of analytics [3]. The widespread use of analytics has made it a key player in the latest societal advancements, facilitating the digitalization of the food industry, and enabling the analysis of food.

A. Big Data

In today's popular publications, whether online or in print, it has become increasingly common to come across references to DS, analytics, BD, or a combination of these terms [5]. BD's size is measured in terabytes or petabytes, with data from various sources like IoT, logs, clickstreams, and social media contributing to its enormity. Historical context from multidimensional data warehouses adds to its complexity. Veracity, a proposed fourth V, highlights data quality, categorized as good, bad, or undefined due to inconsistencies, incompleteness, and ambiguity [6].

Actually, not just corporations and governments generate data; we are all data generators now [7]. We generate data using mobile phones, social network interactions, GPS, etc. Nevertheless, most of such data is not formatted in such a way that can be stored and/ or processed in a standard database management system (DBMS). This necessitates the use of BDA approaches to make sense of such data. However, BD cannot be handled using traditional methods and approaches due to its high volume, variety, velocity, value, and veracity [6].

By making more types of information available and useful at a higher frequency, BD may unlock significant value by increasing product and service development, boosting performance, improving decision-making (DM), and leading to better and more informed management decisions [8].

The systematic extraction of non-obvious and useful patterns and knowledge from data is known as DS [9] [10]. Its goals are to advance research, support organizational DM, and enable a DD society [11]. The definition demonstrates how closely DS is related to science, scientific methodologies, and business analytics. Naturally, much of the definition of science and scientific method, that is, the "*principles and procedures for the systematic pursuit of knowledge involving the recognition and formulation of a problem, the collection of data through observation and experiment, and the formulation and testing of hypotheses*", is shared by DS [12]. As a result, what we designate as data, the means of knowledge extraction, and the end goal distinguish DS [13].

B. Data Science

The use of AI in our daily lives exponentially increases. AI

is the science of programming machines to behave intelligently. The acronym AI is divisive, sparking ongoing philosophical discussions about whether a machine can ever be intelligent. As a result, to escape the controversy surrounding the term AI, many researchers define their work as "intelligent systems" or "knowledge-based systems" [14]. Every day, new AI applications, spanning from ML to robotics and other AI-enabled technology, enter our lives.

At the heart of AI is ML, which is a collection of algorithms used to analyze datasets. By utilizing powerful analytics algorithms, ML strives to process and evaluate large datasets. It enables the development of practical ideas for assessing performance, gaining a competitive edge, and functioning as a new platform for efficiency, creativity, and improved DD DM [15]. The analytics engine, at its core, employs a series of complex algorithms to discover (hidden) patterns in (big) datasets [16].

However, there is a need to bridge the gap between the numerous emerging and very promising gadgets that may be used in the food industry in conjunction with the appropriate data mining and analysis. This multidisciplinary and multidimensional data paradigm will benefit food safety management systems by integrating and crossing several scientific fields and sub-disciplines such as process chemistry development, information technology, food science, food microbiology, molecular biology, process analytical chemistry, vibrational spectroscopy, bioinformatics, ML, chemical engineering, process systems, and control engineering [16].

C. Internet of Things (IoT)

The IoT is a network of physical objects- "things"- embedded with sensors, software, and other technologies for connecting and exchanging data with other devices and systems through the internet. These gadgets range from common domestic items to complex industrial machines. They can communicate with one another and with the physical world by exchanging relevant data between the physical and virtual worlds. These things can also respond autonomously to happenings in their surroundings. All these processes can activate some actions and create services by human intervention or machine-to-machine (M2M) communication [18]. IoT has its application in several areas, such as connected industry 4.0, smart cities, smart homes, smart energy, self-driving cars, smart agriculture, and healthcare.

IoT sensors are used to monitor and measure various farm variables (for example, soil nutrients and weather data) and production factors. Location sensors, optical sensors, mechanical sensors, electrochemical sensors, and airflow sensors are the various types of sensors. Air temperature, soil temperature at various depths, rainfall, leaf wetness, chlorophyll, wind speed, dew point temperature, wind direction, relative humidity, solar radiation, and atmospheric pressure are all measured with these sensors [19].

Sensory devices are inexpensive to operate after the initial expense of purchase and installation, and they create an abundance of data that can be processed for free [20]. Indeed, the application of IoT in industry is sometimes referred to as Industrial IoT (IIoT). Remote sensors receive information

provided by machines in the IIoT framework to boost efficiency, encourage better DM, and establish competitive advantages, independent of industry or company size [21]. Furthermore, it would be imprudent not to use the emerging technologies that collect data, from smart sensors to satellites that can be crunched using various cloud-based analytic software program gear such as AI [22].

IV. FOOD SYSTEM

When discussing the agri-food sector, it's crucial to consider the broader "food system." This section will outline key components, the food ecosystem, consumer trends, food safety, institutional relationships, and the reality of the food context.

A. Food Ecosystem

Humans instinctively are born to thrive for survival. Nowadays, with the increase in famine, capable people intuitively secure themselves with over-buying of food, which results in lots of waste. Food wastes are known to release harmful gases, one of which is methane [23]. As methane production increases, air pollution increases and soil nutrients decrease, which affects crop production, and causes a disturbance in the ecosystem [24]. Hence, controlling crop and food wastes that release harmful gases ought to be addressed for it serves the production efficiency as well as sustainability [23]. Soil organic carbon (SOC) is abundant in soil and is necessary for retaining and improving soil fertility and production. Nowadays, globally cultivated soils have been claimed to have lost 50-70% of their original carbon content. This suggests an ideal chance to return carbon to the soil to elicit a positive response, so improving soil health and addressing a variety of issues such as climate change, food, and water security. BDA has evolved into one of the most potent ways for extracting usable information from data. With geospatial data of very high quality and quantity available, a combination of data assimilation and ML algorithms can also improve SOC dynamic prediction and analysis [25].

B. Food Shortage

As the population continues to grow, the daily struggle to combat hunger becomes more difficult. It is estimated that the global population will reach 10 billion by 2050, leading to an increased demand for food and a decreased availability of resources. A shift to a stronger form of food waste prevention typically requires a significant change to the company's operations and should help them stay afloat for the foreseeable future. Process improvement results in a better fit of food supply and demand, which leads to lower long-term profitability because customers end up buying less. For example, the idea of "zero waste" apps are to make customers at various levels of the supply chain aware of how much food they truly need to buy to meet their needs. The apps recommend quantities that are significantly lower than what clients would have purchased without app support. As a result, these apps decrease the food waste resulting from over-purchasing and spoilage of foods [26]. The agri-food business is a never-ending source of growth for feeding a large population, but there is a

significant need to develop high-standard procedures using intelligent and creative technology like AI and BD.

C. Food Safety

At any point in the farm-to-fork food chain, pollutants that are chemical, biological, or physical can contaminate food. Traditional in-lab safety evaluations are neither appropriate nor practical across the chain of food production and distribution [18]. Thus, a variety of non-destructive methods using sensors have been utilized for identifying hazardous compounds within the food processing and on finished products, these techniques include gas chromatography (GC), high-performance liquid chromatography (HPLC), HPLC-mass spectrometry (HPLC-MS), GC-MS, nuclear magnetic resonance (NMR), fourier transform infrared spectroscopy (FT-IR), and Raman spectrometry. These sensors ought to be strictly used within controlled conditions. Sensors having spectra in the ultraviolet (UV), near-infrared (NIR), mid-infrared, and visible (VIS) ranges have been used in different sensor spectroscopy methodologies to this aim [20].

Due to the expense of herbicides and concerns about their environmental impact, there's increasing interest in using cameras for weed detection in agriculture. Some companies offer autonomous weeding equipment equipped with cameras. Developing commercial harvesters using this technology is vital for labor-saving, productivity, and safety, reducing reliance on harmful chemicals [21].

D. Institutions & Food

Due to processing contaminants, existing and emerging risks, the European Union (EU) has, recently, mandated the use of a mechanism called process analytical technology in the food manufacturing process and production lines. This mechanism addresses a broad demand highlighted by the agri-food industry [20]. Moreover, the World Health Organization (WHO) has lately entailed the BD approach to endorse DM in food safety, which brought about the food safety platform "FOSCOLLAB", which integrates numerous sources from different disciplines. This platform integrates and makes user-accessible dashboards from both structured and unstructured data from a variety of industries, including animal, agricultural, food, public health, and economic indicators [27].

The Food, Energy, Water (FEW) nexus, a UN paradigm, highlights the interconnectedness of food, energy, and water systems. Changes in one area impact others. FEW research requires innovative methodologies to address complex interdependencies over space and time. To establish sustainability, the FEW nexus identifies overlap or conflict among system components, aiming to enhance efficiency, eliminate trade-offs, foster synergies, and improve governance. These insights inform models of ecosystem service flows and trade-offs, guiding policy decisions [1].

V. FOOD ANALYTICS

A. Big Data in Food Analytics

Since 2013, BD has revolutionized food safety, security, and agriculture. It's reshaping supply chains, with data sources

including regulatory, food enterprise, and media data. Quality BDA can drive success, but poor data quality hampers market forecasting and social stability [28].

The recent introduction of advanced technologies has changed practitioners' and academics' attention to the deployment of enormous amounts of created data as a critical and efficient instrument for addressing modern food supply chain management concerns. Surprisingly, the annual added value that could be realized using BD in food supply chains is projected to be between \$120 and \$150 billion [29]. BD is mostly used in observing the concerns controlling global problems such as food security, safety, feasibility, and growth in efficiency, which expands the scope of BD beyond farming and includes the entire food supply chain. The interconnection of agricultural and supply chain devices results in real-time data evolution via hassle-free wireless IoT [30].

Previous literature shows that apart from revenue growth and cost optimization, DA can decrease customer acquisitions costs, retain valuable customers, help predict customer behavior, improve customer experience, reduce fraud, provide real-time offers, and enhance DM. However, data on its own is not a source of competitive advantage since all firms can collect hordes of data from a variety of sources. Rather, data must be purposely analyzed, and activated [31]. According to reference [32], BD emphasizes "utility from" data rather than "ownership of" data. This means that access to purposeful data is key. Further, raw data is useless unless it is purposefully analyzed.

To conclude, information and insights can be derived from BD by applying advanced analytics (i.e., BDA), which brings unprecedented opportunities to benefit from it. In fact, previous literature has found that firms using BDA are 36% more likely to surpass their competitors in revenue growth, operating efficiency, and can decrease their customer acquisition costs by 47% [31].

B. Data Science in Food Analytics

BDA can be used to improve sustainability performance of supply chains [29]. Studies also demonstrate how BD collected along all phases of agricultural supply chains, from pre-production to distribution, and analyzed using ML approaches can improve supply chain performance and environmental sustainability. For instance, accurate weather forecasts can help with water resource management. Furthermore, the capacity to extract meaningful insights from BD may increase supply chain visibility and facilitate collaboration for long-term sustainability. In general, supply chain environmental sustainability can be improved by employing BDA to estimate the environmental impact of supply chain-related choices [33].

Climate change, population growth, water shortages, soil degradation, and food security are all important concerns confronting traditional agriculture [1][23]. The data collected from sensors, such as the pH sensor and the temperature sensor, is processed using ML techniques to determine the system's health. Countries with a diverse land distribution and scarcity of water for home and agriculture use require a viable solution to food and environmental issues. The Aquaponics farming method approach promises to be a prominent alternative. For

effective growth and information usage, cloud-based sustainable smart aquaponics farming employs IoT-based predictive analytics. Aquaponics is a closed-loop symbiotic system that allows aquatic life and plants to coexist. Aquatic animals produce a nutrient-rich byproduct (ammonia), which plants utilize as fertilizer. This system involves several mechanical and biological operations like heating, pumping, filtering, and so on [22]. DA and sensor technology enable the creation of optimal conditions for both plants and fish, ensuring high output and improving resource efficiency by regularly monitoring the water quality, water level, temperature levels, fish health, salinity, pH level, humidity, and sunlight. Accurate data can boost fish and plant yields by allowing the nutrients transfer between plants and fish. The data can also be utilized to automate tasks that require less human participation [19].

C. Artificial Intelligence & Machine Learning in Food Analytics

AI's significance has been deemed deserving though poorly implemented. Monitoring optimum environmental settings avails the world's growing population. AI helps in detection using sensors and sets optimum farming conditions, for greater and better crop yields, which is critical to meeting the needs of the growing inhabitants of the world. In addition, BDA helps farmers track down problems early and hence take necessary constructive and/ or preventive actions, to enhance production [23].

BDA and allied technologies in food science and related domains improve performance through IoT for food safety and supply chain traceability to solve food security challenges. Some AI developments in the supply chain are AI-based demand forecasting, risk-management, resilience, transportation, supplier selection, inventory management, etc. Automation and intelligent systems can also be utilized to choose appropriate chemicals for food safety, and multitasking robots can help to speed up the process while maintaining quality [30].

The food systems perspective has become increasingly important as meeting the targets of SDG 2 to end hunger, achieve food and nutrition security, improve nutrition, and promote sustainable agriculture necessitates more holistic approaches to understanding food insecurity and nutrition deficiencies that critically analyze interactions between food production and consumption and their socioeconomic and institutional contexts. One significant challenge is exploiting integrated data that can talk about the growth of food systems and the interconnections between their components [34].

D. IoT in Food Analytics

In the agricultural sector, IoT and DA are employed to improve operational efficiency and productivity. The usage of WSN as a main driver of smart agriculture is giving way to the use of IoT and DA. To meet the world's food demands in the next years, the globe is adopting the use of IoT paired with DA. Smart agriculture will be possible using IoT and DA, which is predicted to produce great operational efficiency and yield [19].

There are numerous instances of IoT use in agriculture [21].

Crop and livestock tracking, machinery, irrigation, weather monitoring, soil monitoring, disease and pest management, automation, greenhouse production, and precision agriculture are examples. Environmental conditions like rainfall, temperature, humidity, soil moisture, salinity, and pest migration influence crop yield. Acquiring such data aids in understanding farm processes. It enables optimal data management to improve produce quality, reduce risks, and optimize profitability [19].

Smart farming enables farmers and growers to increase output while decreasing waste, from the amount of fertilizer used to the number of trips made by farm vehicles. Data may be accessed whenever and wherever by connecting IoT devices to the Internet. However, data transfer through the internet necessitates security measures, real-time data support, and flexibility of access. The internet managed to clear the path for cloud computing, which entails managing user interfaces, services, organizing and coordinating network nodes, computation, and data processing. The usage of cloud IoT platforms enables the storage of large amounts of data collected from sensors in the cloud. However, there are several critical technical criteria to consider while deploying IoT devices. The following factors should be examined for wireless connectivity: communication distance, data rate, battery life, mobility, latency, security and resilience, and gateway modem cost [19].

To conclude, BDA assists companies in the food industry to understand customers' needs and preferences by gathering real-time information about consumers, products, brands, and competitors. As a result, it plays a vital role in the expansion of the food industry. DA is a complicated process where raw data is translated into meaningful insights, that drives business decisions. It is just a matter of employing the proper tools and understanding what to do with the gathered data [35]. For this, decision-makers leverage new technologies to further support their businesses. In [36], the author claims that DA unlocks great value by increasing the employees' efficiency, reducing waste, ensuring consistent quality, optimizing product portfolios, and finally becoming a DD organization.

VI. RESEARCH AGENDA

The literature review synthesis guides further analysis of future research directions in BD, analytics, and food convergence, forming food analytics. Studies highlight adoption barriers in food and farming sectors utilizing BD and analytics algorithms, including trust, data privacy, complexity, lack of IoT knowledge, and high costs. This prompts essential research questions, such as:

- *What risks may smart food production using BDA create for users in terms of data privacy and security? How to overcome these adoption barriers?*
- *How can BDA contribute to building trust for food industry users to increase the adoption of smart food and farming solutions (e.g., use of IoT sensors)?*

The increased sharing and exchange of data poses a significant challenge to the EU agri-food sector. Agriculture data has a very specialized and diverse nature (e.g., livestock

and fish stock data, land and agronomic data, climate data, machine data, financial data, compliance data etc.). When we consider the entire supply chain, the problem gets much more confusing, because information about food origin, quality, and safety, including nutritional information, is mixed up with information about production, energy, distribution, and waste management. So, we need to address:

- *How to reap all the potential benefits of data sharing between many stakeholders? How to ensure a fair and transparent regulations?*
- *How to ensure that the code of conduct for data sharing in agriculture provides the prerequisites for defining the soft infrastructure based on contractual agreements, as well as guidelines on fair and transparent data use?*

One critical problem is whether farm management system (FMS) manufacturers are willing to allow interoperability and federation of their data platforms with other systems. So, there is a need-to-know answer to several questions, such as:

- *How to enable interoperability and data platforms federation of the food industry and ecosystem?*
- *What are, and how to implement, a set of design principles allowing the creation and maintenance of data space in food industry where data is governed, managed, and exchanged seamlessly and safely?*

Even though some studies have discussed utilization of AI techniques and the intelligent optimization algorithm for food production management, further research is necessary to address these kinds of managerial aspects, when food industry benefits from AI based solutions. Therefore, the question that should be answered is:

- *How and in what ways BDA and more particularly AI based techniques may contribute to the smart food industry in managerial level (concerning the processes and so on)?*

Some of the reviewed studies highlighted the fact that using BDA and smart solutions such as IoT and AI technologies may affect the quality of the produced food. This in turn will affect the quality of humans' life in the long run. Therefore,

- *What are the direct and indirect effects of using BDA on quality of life for humans in a long-term period?*

When it comes to environmental sustainability, several studies have highlighted the importance of applying BDA in the food industry. However, these articles have not thoroughly investigated this aspect and further research is required in this regard. For example,

- *In what ways does employing BDA-based techniques such as ML, IoT, or cloud-based analytics in the food industry affect environmental sustainability?*

Various studies have explored food situation prediction, covering aspects like customer behavior and risk assessment. While much attention is on the "why," there's a gap in understanding how these predictions inform decision-making in food industries. Key questions arise:

- *How the results of applying BDA in food industry may help us to make DDD to mitigate risks in food industry*
- *How IoT-based predictive analytics in the food industry may empower DD DM?*

Agriculture is entering a digitally enhanced farming era, with data generated at various phases of agricultural production and related operations, and IoT solutions, robotics, and sophisticated decision-support systems being used. DD approaches provide unprecedented opportunities to 1) improve resource efficiency, productivity, and environmental sustainability; 2) dynamically adapt business plans to changing markets and consumer expectations; 3) reduce administrative costs and enable science-based policies; 4) provide more prosperous living conditions for rural communities; and 5) improve the consumer's relationship with the various actors across the value chain. This study provides a comprehensive review of the application of BD, DS, AI, ML algorithms, and IoT in the food industry, and discovers areas which require further research efforts and investigation, as well as posing questions for researchers, practitioners and governmental food agencies and officials, with regards to the technical challenges, the privacy and security concerns, and the legislative restraints. The review conducted in this study is invaluable in helping to advance the food industry, and its findings could be used to inform future developments in the industry, in a cross-disciplinary fashion.

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